

02LQD Specification and Simulation of Digital Systems

2020/01/31

1. Combinational circuit design (3 points)

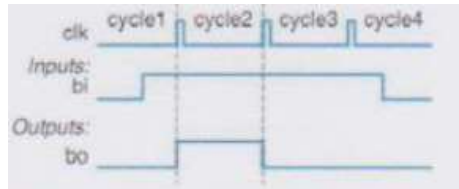
A network router connects multiple computers together and allows them to send messages to each other. If two or more computers send messages simultaneously, the messages collide and must be resent. Design a combinational collision detection circuit with 4 inputs M_0 , M_1 , M_2 , M_3 that are 1 when the corresponding computer is sending a message, 0 otherwise. The output C is 1 when a collision is detected, 0 otherwise.

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (12 points)

Design a Moore FSM that synchronizes a button press to a clock signal, such that, when the button is pressed (the 1-bit input signal b_i is 1), the output signal b_o is 1 for exactly one clock cycle. The b_o synchronized output is used to prevent a single button press that lasts for multiple clock cycles from being interpreted as multiple button presses. The circuit waits for b_i to return to 0 and then restarts operation waiting for b_i to become 1 again, indicating that the button has been pressed once more. The following figure shows an example of the FSM's timing.



Reset is synchronous.

1. Draw the state diagram (4 points)
2. Design the FSM resorting to the VHDL behavioural description style with 2 processes (4 points).
3. Design the FSM resorting in Verilog with 2 always blocks (4 points).

3. FSM-D design (15 points)

Design a FSM-D for an alarm system. At each clock cycle the 1-bit output `alarm` is set to 1 whenever the average of the last 4 consecutive samples exceeds a threshold, otherwise it is set to 0. The first input is a 16-bit unsigned binary number `Temperature` for the physical quantity, e.g. the temperature. The second input is a 16-bit unsigned binary number `Threshold` for the warning threshold. Reset is asynchronous.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (7 points)
2. Design a FSM-D (8 points):
 - create 2 processes for behavioural datapath description from the HLSM state diagram
 - create 2 processes for behavioural controller description from the HLSM state diagram
 - connect datapath processes to controller processes to create a single architecture.

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
 - running VHDL code and testbench
 - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam
- **no later than Monday February 3 11.59 pm**
- format of .zip archive name surname_ID (example Rossi_244123)
- remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.